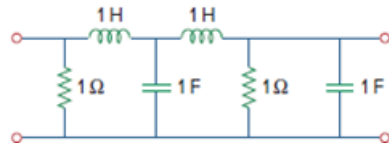


Problem

Determine the **ABCD** parameters of the circuit in Fig. 19.120 as functions of s . (*Hint*: Partition the circuit into subcircuits and cascade them using the results of Prob. 19.43.)

Figure 19.120



Step-by-step solution

Step 1 of 6

Draw the following circuit diagram.

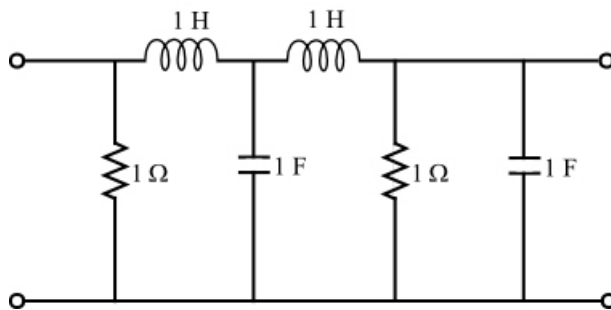


Figure 1

Step 2 of 6

Divide Figure 1 into sequence of single elements which are as shown below.

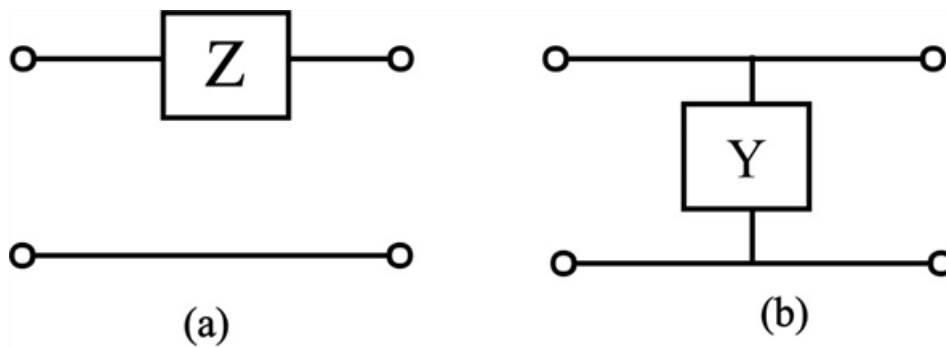


Figure 2

Step 3 of 6

From Figure 2, the transmission parameters for are as follows.

$$[T_a] = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

$$[T_b] = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z} & 0 \end{bmatrix}$$

Where,

$[T]$ is transmission matrix.

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Step 4 of 6

Divide Figure 1 into six sub circuits which are as shown below.

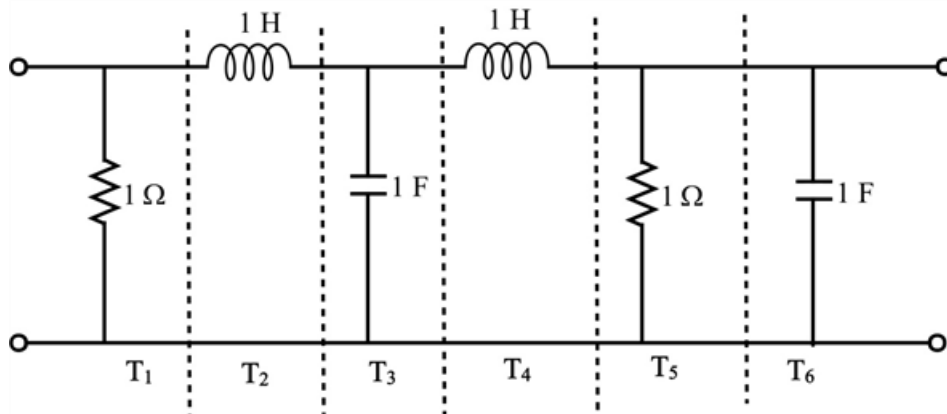


Figure 3

Step 5 of 6

Modify Figure 3 into Laplace domain which is as shown below.

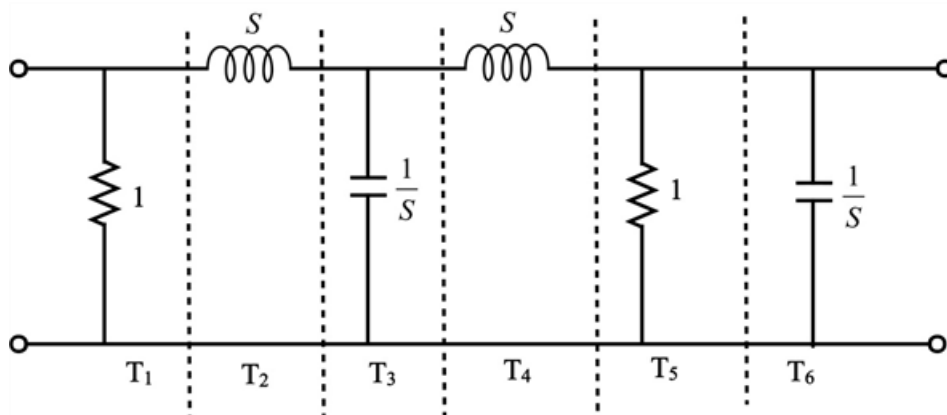


Figure 4

Step 6 of 6

From Figure 4, write the transmission matrix for single element.

$$[T_1] = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$[T_2] = \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix}$$

$$[T_3] = \begin{bmatrix} 1 & 0 \\ S & 1 \end{bmatrix}$$

$$[T_4] = [T_2]$$

$$[T_5] = [T_1]$$

$$[T_6] = [T_3]$$

Write the overall transmission matrix.

$$[T] = [T_1][T_2][T_3][T_4][T_5][T_6]$$

Substitute the matrices of $T_1, T_2, T_3, T_4, T_5, T_6$ in $[T] = [T_1][T_2][T_3][T_4][T_5][T_6]$.

$$\begin{aligned} [T] &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ S & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ S & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ S & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ S+1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ S & 1 \end{bmatrix} \begin{bmatrix} 1+S^2+S & S \\ S+1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & S \\ 0 & 1 \end{bmatrix} \begin{bmatrix} S^2+S+1 & S \\ S^3+S^2+2S+1 & S^2+1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} S^4+S^3+3S^2+2S+1 & S^3+2S \\ S^3+S^2+2S+1 & S^2+1 \end{bmatrix} \\ &= \begin{bmatrix} S^4+S^3+3S^2+2S+1 & S^3+2S \\ S^4+2S^3+4S^2+4S+2 & S^3+S^2+2S+1 \end{bmatrix} \end{aligned}$$

Therefore, the **ABCD** parameters of the circuit are:

$$\begin{bmatrix} S^4+S^3+3S^2+2S+1 & S^3+2S \\ S^4+2S^3+4S^2+4S+2 & S^3+S^2+2S+1 \end{bmatrix}$$